Intermediate Coding Practice Questions

Find the Largest Prime Factor

Given a positive integer `n`, find its largest prime factor.
You need to find an efficient solution (less than O(n)).

Input Format

Input: A positive integer n.

Output Format

Output: The largest prime factor of the given number.

Power of a Number (Recursion)

Write a function that calculates `x^y` (x raised to the power y) using recursion.
The function should work for both positive and negative exponents.

Input Format

Input: Two integers x and y.

Output Format

Output: x raised to the power y (x^y).

Find the GCD (Greatest Common Divisor) of Two Numbers

Write a function that finds the greatest common divisor (GCD) of two numbers using the Euclidean algorithm.

Input Format

Input: Two integers a and b.

Output Format

Output: The greatest common divisor of a and b.

Tower of Hanoi Problem

Write a recursive solution to the Tower of Hanoi problem, where you need to move `n` disks from a source pole to a destination pole, using a temporary pole.
You can only move one disk at a time, and no disk may be placed on top of a smaller disk.

Input Format

Input: An integer n representing the number of disks.

Output Format

Output: The sequence of moves to transfer the disks from the source pole to the destination pole.

Sum of Multiples of 3 or 5 Below N

Find the sum of all multiples of 3 or 5 below a given number `n`.

Input Format

Input: A positive integer n.

Output Format

Output: The sum of all multiples of 3 or 5 below n.

Find the Nth Fibonacci Number Using Recursion

Write a recursive function to find the Nth Fibonacci number.
The function should return the Nth number in the Fibonacci series.

Input Format

Input: A positive integer n representing the position in the Fibonacci sequence.

Output Format

Output: The nth Fibonacci number.

Check if a Number is a Perfect Number

Write a function that checks if a given number is a perfect number. A perfect number is a positive integer that is equal to the sum of its proper divisors (excluding the number itself).

Input Format

Input: A positive integer n.

Output Format

Output: True if n is a perfect number, otherwise False.

Find the Least Common Multiple (LCM)

Write a function to calculate the least common multiple (LCM) of two numbers.
Use the formula:
`LCM(a, b) = (|a * b|) / GCD(a, b)`

Input Format

Input: Two integers a and b.

Output Format

Output: The least common multiple of a and b.